Code No: 138DU **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, September - 2022
NEURAL NETWORKS AND DEEP LEARNING
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) What is perceptron? Explain the working of the perceptron.
b) Give an overview of Hopfield network. [8+7]

2. Explain about back propagation algorithm for training a neural network. [15]

3.a) Describe the architecture of max net.
b) What are the salient features of counter propagation network? [8+7]

4. Explain how Kohonen self-organizing feature maps work. [15]

5.a) Give the historical evolution of neural networks and deep learning.
b) Illustrate the learning of XOR function using deep feed-forward network. [7+8]

6.a) Describe the various choices for cost functions in neural networks trained with gradient-based optimization?
b) What architectural considerations are important in neural networks? [8+7]

7. What do you mean by regularization? Explain any two regularization techniques. [15]

8.a) Discuss with stochastic gradient-descent algorithm.
b) Explain the use of deep learning in speech recognition. [8+7]

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